Assignment 1 (Lecture 2): House Price Prediction via Linear Regression

Topics: Optimization, Data Normalization, Closed-Form Solutions, Model Evaluation

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Objective

In this assignment, you will build, evaluate, and compare multiple linear regression approaches to predict house prices using the provided dataset. Your goal is to demonstrate proficiency in algorithm implementation, data visualization, empirical evaluation, and critical discussion of machine learning fundamentals.

Dataset Setup

- **Dataset:** Available on Kaggle via Housing Prices Prediction using Linear Regression.
- **Data Splitting:** The dataset is provided as a single set. You must perform an 80/20 train/test split (`train` $= 0.8$, `test` $= 0.2$) before training or evaluating any models.
- **Feature Selection:** The target variable (y) is the house `Price`. You are free to select and engineer the input features (X) you deem most predictive.

Task Instructions

You are required to implement and evaluate linear regression using three distinct approaches:

1. **Method A:** Gradient descent (GD), without feature scaling
2. **Method B:** Gradient descent with input data normalization
3. **Method C:** Closed-form solution (normal equation)

Evaluation Criteria & Required Deliverables

Your submission will be evaluated based on the following four key pillars:

1. Implementation

- Correct implementation of the loss function (MSE), gradient updates, and closed-form math from scratch (or standard library foundations).
- Proper handling of data preprocessing, feature selection, and data splitting without data leakage.

2. Visualization

- **Convergence curves:** Plot Training Loss vs. Iterations for Methods A and B to show model learning trajectories.
- **Prediction vs. ground truth:** Scatter plots comparing predicted house prices against actual values on the test set across all three methods.

3. Evaluation

- Evaluate performance on both training and test sets using standard regression metrics: Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R^2 Score.
- **Evaluation metrics:** Present your findings using both quantitative performance metrics (summarized cleanly in a comparative table) and qualitative analysis (evaluating error patterns and model behaviors across the test set).

4. Discussion & Critical Analysis

- **Feature selection justification:** Explain the rationale behind your chosen input features and how they influenced model performance.
- **Normalization impact:** Analyze the effect of input data normalization on the convergence rate and stability of GD.
- **Closed-form vs. iterative:** Compare the computational efficiency and output accuracy of the close-form against GD.